

Lecture 18: Model stealing

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1 Model Stealing for Any Low-Rank Language Model

Based on *Model Stealing for Any Low-Rank Language Model* [Liu and Moitra, 2025], presented by Tanmay Sinha.

1.1

“Model stealing, where a learner tries to recover an unknown model via carefully chosen queries, is a critical problem in machine learning, as it threatens the security of proprietary models and the privacy of data they are trained on... our algorithm succeeds at stealing any language model whose output distribution is low-rank.”

No assumptions about how the model is trained. We’ll treat it as a black box.

$O \rightarrow$ Finite alphabet

$T \rightarrow$ length of string

$H \rightarrow$ A language model (distribution over strings of length T)

Assume that H is low-rank:

$$\forall t \in [T], P(f|h) \in \mathbb{R}^{O^{T-t} \times O^t} \text{ is low rank (has rank } \leq S)$$

To clarify: “Low-rank” here does not indicate the usage of “Low-rank Adaptation (LoRA)”. It refers to the rank of the probability matrix relating prompts and responses.

If you can find efficient representations of S of those bases (columns), then you can efficiently sample from it.

Conditional Query Model: For any $h \in O^t$, you can sample $f \sim Pr[\cdot|h]$

Theorem 1. *With $\text{poly}(S, |O|, T, 1/\eta)$ queries and $\text{poly}(S, |O|, T, 1/\eta)$ time, you can learn a distribution $\hat{H}$ such that:*

(1) $d_{TV}(\hat{H}, H) \leq \eta$ where d_{TV} is total variation distance, a measure of distance between distributions.

(2) You can sample from $\hat{H}$.

Claim 2. Assume, for each $t \leq T$, you have the following:

- $\mathcal{H}^{(t)} \subseteq O^t, |\mathcal{H}^{(t)}| \leq S \wedge \{Pr[\cdot|h]\}_{h \in \mathcal{H}^{(t)}}$ s.t. it spans $\{Pr[\cdot|h]\}_{h \in O^t}$
- For each $h \in \mathcal{H}^{(t)}$, next character probability is $Pr[o|h]$
- (Change-of-basis) $\forall h \in \mathcal{H}^{(t)}, o \in O, \{\alpha_{h'}^{\text{cat}(h,o)}\}_{h' \in \mathcal{H}^{(t+1)}}, Pr[\cdot|\text{cat}(h,o)] = \sum_{h' \in \mathcal{H}^{(t+1)}} \alpha_{h'}^{\text{cat}(h,o)} Pr[\cdot|h']$

Then we can sample from $\mathcal{H}$

Proof. Let β represent the coefficients of the bases.

$$x = o_1 o_2 \dots o_t$$

$$Pr[\cdot|x] = \sum_{h \in \mathcal{H}} \beta_x^h Pr[\cdot|h]$$

$$\text{Sample: } \sigma_{t+1} \sim Pr[o|x] = \sum_{h \in \mathcal{H}^{(t)}} \beta_x^h Pr[o|h]$$

Where $Pr[o|h]$ comes from assumption 2

$$\begin{aligned} \text{Renormalize: } Pr[\cdot|\text{cat}(h,o)] &= \sum_{h \in \mathcal{H}^{(t)}} \beta_x^h \frac{Pr[o|h]}{Pr[o|x]} Pr[\cdot|\text{cat}(h,o)] \\ &= \sum_{h' \in \mathcal{H}^{(t+1)}} \left(\sum_{h \in \mathcal{H}} \beta_x^h \alpha_{h'}^{\text{cat}(h,o)} \frac{Pr[o|h]}{Pr[o|x]} Pr[\cdot|h'] \right) \end{aligned}$$

For each h , store the empirical probability vectors. □

1.2 Discussion

The coefficients are found based on next-token probabilities.

Why low-rank? Before ML-based language models, the prior Markov models are low rank. Additionally, language models must be simple in “some sense” since they are learnable from data. There is a lot of work going into finding where this regularization/simplicity is happening. Recent work suggests that the log probability matrices might be low-rank.

References

Allen Liu and Ankur Moitra. Model stealing for any low-rank language model. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing*, STOC '25, page 1755–1761, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400715105. doi: 10.1145/3717823.3718220. URL <https://arxiv.org/abs/2411.07536>.

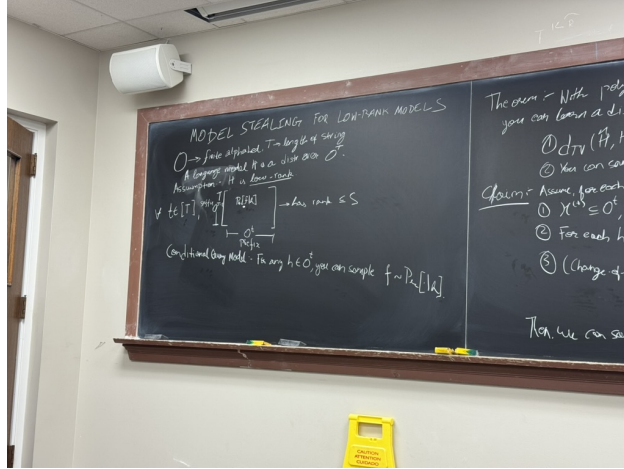


Figure 1: Blackboard 1

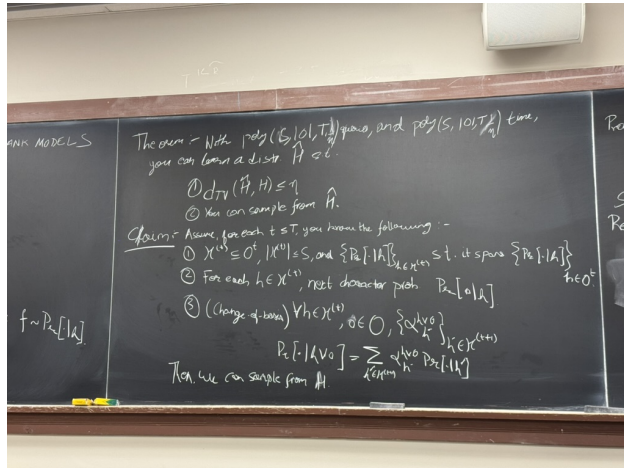


Figure 2: Blackboard 2

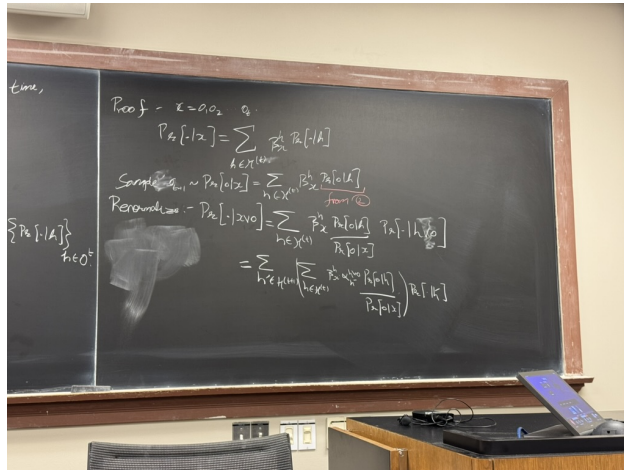


Figure 3: Blackboard 3